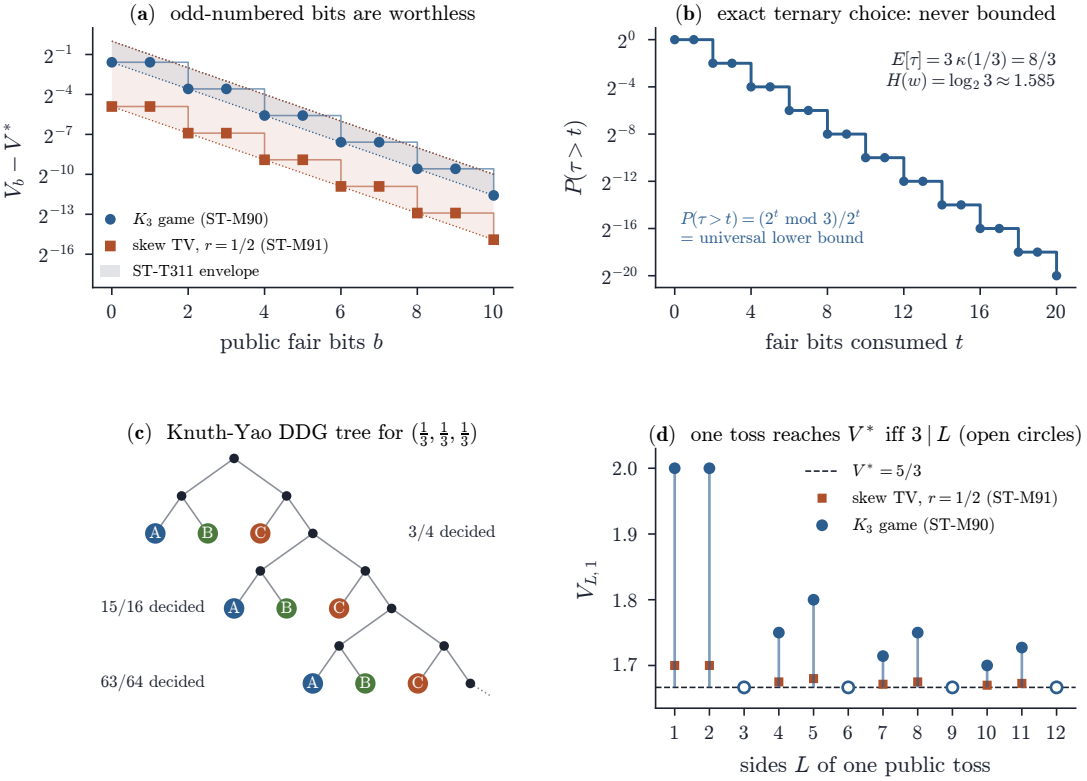
(a) odd-numbered bits are worthless
$V_b - V^*$
2^{-1}
2^{-4}
2^{-7}
2^{-10}
2^{-13}
2^{-16}
K_3 game (ST-M90)
skew TV, $r = 1/2$ (ST-M91)
ST-T311 envelope
0
2
4
6
8
10
public fair bits b
(b) exact ternary choice: never bounded
$P(\tau > t)$
2^{0}
2^{-4}
2^{-8}
2^{-12}
2^{-16}
2^{-20}
$E[\tau] = 3\,\kappa(1/3) = 8/3$
$H(w) = \log_2 3 \approx 1.585$
$P(\tau > t) = (2^t \bmod 3)/2^t$
= universal lower bound
0
4
8
12
16
20
fair bits consumed t
(c) Knuth-Yao DDG tree for $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
A
B
C
3/4 decided
15/16 decided
63/64 decided
(d) one toss reaches V^* iff $3 \mid L$ (open circles)
$V_{L,1}$
2.0
1.9
1.8
1.7
$V^* = 5/3$
skew TV, $r = 1/2$ (ST-M91)
K_3 game (ST-M90)
1
2
3
4
5
6
7
8
9
10
11
12
sides L of one public toss